# COMPLEX STRUCTURES ON $\mathbb{S}^6$

PHILIP ENGEL

Abstract. A recent manuscript produced by Claude under the direction of Levent Alpöge constructs a 1-parameter family of compact complex threefolds diffeomorphic to $\mathbb{S}^6$ [AC26]. Our goal is to describe the geometric ideas behind the construction, give a self-contained proof, and remark on possible future research directions.

Contents



## 1. Introduction

1.1. **History.** In 1948, Hopf proved that $\mathbb{S}^4$ and $\mathbb{S}^8$ do not admit complex structures [Hop48] and in 1953, Borel and Serre [BS53] used topological methods to prove that only $\mathbb{S}^2$ and $\mathbb{S}^6$ admit an almost complex structure. Since $\mathbb{S}^2 \simeq \mathbb{P}^1$, this raises the question:

**Question 1.1** (Hopf Problem)**.** *Does $\mathbb{S}^6$ admit a complex structure?*

Yau asked whether every compact almost complex manifold of dimension at least 6 also admits a complex structure [Yau93, Prob. 52].

Both negative and positive answers to Question 1.1 have been proposed, but these works were found not to be fully mathematically sound, see [ABGKR18] and [Bry21] for discussions. Thus, the problem acquired a certain degree of notoriety. Hypothetical properties of such a complex structure, for instance, Dolbeault cohomology groups and algebraic dimension, were studied in [Gra97], [CDP98].

In 2026, L. Alpöge[1] posted a 108-page manuscript [AC26] produced by an artificial intelligence, Anthropic's "Claude", claiming to construct a complex structure on $\mathbb{S}^6$. Perhaps the most remarkable aspect of the construction is that it fibers holomorphically over the Riemann sphere $\mathbb{P}^1$, contradicting a nearly 30-year-old result in the literature that the algebraic dimension of such a complex structure must be zero [CDP98, Cor. 4.2]. See [AC26, § 10] for a further discussion.

1.2. **Outline.** We begin with the rational elliptic surface $S$ having singular fibers of Kodaira types $IV^*$, $III$, $I_1$. From a line bundle $M \in \mathrm{Pic}(S)$, one forms a principal $\mathbb{G}_m$-bundle over $S$, then quotients it by a lift of translation along a section of $S \to \mathbb{P}^1$. This linearization acquires a zero along $\infty \in \mathbb{P}^1$ where one fills the quotient family by a Mumford construction. The result is a compact complex 3-fold $X(u)$ fibering over $\mathbb{P}^1$, with complex 2-torus fibers away from $0, 1, \infty$. It depends on a single complex parameter $u \in \Delta^*$. See Section 2.

Using the potential good reduction of $S$ over $0, 1 \in \mathbb{P}^1$ to perform logarithmic transforms, one can modify the family over $0, 1$. The replaced fibers have multiplicities $3, 4$ respectively, and their reductions are bielliptic surfaces. The result is a compact complex 3-fold $Y = Y(u)$ fibering over $\mathbb{P}^1$. See Section 3. The monodromy representation of the fibration $Y \to \mathbb{P}^1$ is computed in Section 5. Using this, one checks by van Kampen that $Y$ is simply connected (Section 6), and by the Leray spectral sequence, that $Y$ is a $\mathbb{Z}$-homology sphere (Section 7). Hence $Y$ is diffeomorphic to $\mathbb{S}^6$ (Section 4).

We speculate about possible future research directions in Section 8.

1.3. **Relation to the original manuscript.** In [AC26], the geometry of the construction is somewhat obscured. Rather, the periods of the complex 2-tori are the central focus, encoded by a period matrix

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix}.$$

Here $z$ is a coordinate on the universal cover of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Roughly, the relation to this paper is that $\tau$ is the period of the elliptic fibration $S \to \mathbb{P}^1$, $\mu$ encodes the third period arising from the circle of the principal $\mathbb{G}_m$-bundle, and $\beta$ encodes the fourth period, from the quotient by the $\mathbb{Z}$-action of the lifted translation.

[1] The precise nature of the interaction between Alpöge and Claude is not, at present, documented.

This paper recasts several arguments in [AC26], such as the van Kampen and Leray computations. Since our geometric approach differs substantially from a period-theoretic presentation, we give a self-contained account of the construction and proof. We have omitted some material from the original text, such as a detailed discussion of [CDP98] and computations of Dolbeault cohomology groups.

1.4. **AI disclosure.** The exploration of the original paper and several of the computations were performed in collaboration with ChatGPT 5.6-Sol. The mathematical content was independently verified by the author, who takes full responsibility for the manuscript below.

## 2. Initial construction

Let $\pi : S \to \mathbb{P}^1$ be a rational elliptic surface whose $j$-map has degree 1, so that in suitable coordinates, $j = 1728t$, and for which there are three singular fibers, of Kodaira types $IV^*$, $III$, $I_1$ [Mir90, p. 206]. See Figure 1. The singular fibers correspond respectively to the order 3, 4, $\infty$ corners of the $(3, 4, \infty)$ triangle. Concretely,

$$y^2 = x^3 - 3t^3(t-1)x + 2t^4(t-1)^2.$$

There is a zero section $O$, and the Mordell–Weil group is infinite cyclic [OS91, Main Thm., no. 49]. Let $P \in \mathrm{MW}(S)$ be a generator. The section $P$ passes through non-identity components on both the $IV^*$ and $III$ fibers, whose component groups are $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ respectively.

Consider the line bundle $M = \mathcal{O}_S(P - O)$. Translation by $6P$ defines an automorphism $t_{6P} : S \to S$. Since 2 and 3 both divide 6, the automorphism $t_{6P}$ acts in a manner preserving the components of the $IV^*$ and $III$ Kodaira fibers.

**Lemma 2.1.** $\mathcal{H}om(t_{6P}^* M, M) \simeq \pi^*\mathcal{O}_{\mathbb{P}^1}(1)$.

*Proof.* For a general fiber $S_t$ over $t \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$, we have that $M_t \in \mathrm{Pic}^0(S_t)$. It follows that $t_{6P}^* M_t \simeq M_t$ for all $t \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$. Hence $t_{6P}^* M \otimes M^{-1}$ is expressible as the line bundle associated to a linear combination of vertical divisors. Since $t_{6P}$ acts by an element of $\mathrm{Aut}^0$ of every fiber, the line bundle $L := \mathcal{H}om(t_{6P}^* M, M)$ has multidegree 0 on every fiber. Thus, $L \simeq \pi^*\mathcal{O}_{\mathbb{P}^1}(n)$ for some integer $n$, seeing as $\pi$ has no multiple fibers.

We compute $n$ via Shioda's height pairing. Since $S$ is rational elliptic, we have $\chi(\mathcal{O}_S) = 1$, $O^2 = -1$, and $P \cdot O = 0$. The local contributions at the $IV^*$ and $III$ fibers are $\frac{4}{3}$ and $\frac{1}{2}$, respectively [Shi90, Thm. 8.6 and Table 8.9]. Hence

$$\langle P, P \rangle = 2 - \tfrac{4}{3} - \tfrac{1}{2} = \tfrac{1}{6}.$$

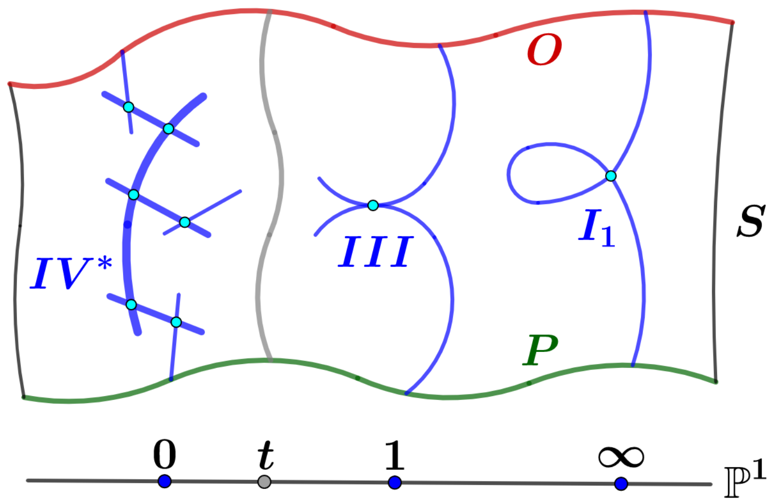


FIGURE 1. Rational elliptic surface $S$ with $IV^*$, $III$, $I_1$ singular fibers in blue (respectively: $\widetilde{E}_6$ configuration, tacnode, and irreducible nodal curve). Origin section $O$ in red, generator $P$ of the Mordell-Weil group in green.

Consider the section $Q = 6P$. By bilinearity,

$$\langle Q, Q\rangle = 36\langle P, P\rangle = 6.$$

Since $Q$ meets the identity component of every reducible fiber, all local contributions involving $Q$ vanish, and Shioda's height formula gives

$$6 = \langle Q, Q\rangle = 2 + 2(Q \cdot O).$$

It follows that $(6P) \cdot O = Q \cdot O = 2$. Similarly, $\langle P, Q\rangle = 6\langle P, P\rangle = 1$ and in turn $P \cdot (6P) = P \cdot Q = 2$.

Finally, restricting $L = t_{6P}^* M^{-1} \otimes M$ to the zero section gives

$$\begin{aligned} n &= L \cdot O \\ &= (O - P) \cdot (6P) - (O - P) \cdot O \\ &= (2 - 2) - ((-1) - 0) = 1. \end{aligned}$$

□

**Corollary 2.2.** *There is, up to scale, a unique section*

$$\phi \in \mathcal{H}om(t_{6P}^* M, M)(S)$$

*which over* $\mathbb{A}^1$ *is an isomorphism, i.e. a linearization of the action of* $t_{6P}$ *on* $M$*, and having a simple zero along the fiber over* $\infty \in \mathbb{P}^1$*.*

Let $M^\times := M \setminus \{\text{zero section}\}$. It is a threefold, which is a principal $\mathbb{G}_m$-bundle over the rational elliptic surface $S$. Let $S^\circ := S \setminus S_\infty$ and $M^{\times,\circ} := M^\times|_{S^\circ}$.

**Definition 2.3.** We define a $\mathbb{Z}$-action on $M^{\times,\circ}$ by acting by the translation $(t_{6P})^{-1}$ on the base $S^\circ$ and the linearizing isomorphism $\phi$ on $M^{\times,\circ}$. That is, the generator of $\mathbb{Z}$ acts by the lift of $t_{6P}^{-1}$ defined by $\phi$.

So in particular, the automorphism of $M^{\times,\circ}$ induced by $\phi$ lifts the automorphism $t_{6P}^{-1}$ of $S^\circ$.

**Proposition 2.4.** *Fix a reference linearization $\phi$ as in Corollary 2.2. There exists a constant $c$ such that for all linearizations $u\phi$, $u \in \mathbb{C}^*$, with $|u| < c$, the $\mathbb{Z}$-action of Definition 2.3 is properly discontinuous.*

*Proof.* Consider the map $\phi\colon t_{6P}^*M \to M$ which is an isomorphism away from $S_\infty$. Choose a smooth Hermitian metric on $M$. With respect to this metric, the operator norm $x \mapsto \|\phi_x\|$ is a continuous function on the compact surface $S$, equal to zero exactly along $S_\infty$. It is therefore bounded. Set $C = \max_{x\in S} \|\phi_x\|$ and choose $c > 0$ such that $cC < 1$.

Fix $u \in \mathbb{C}^*$ with $|u| < c$. Then, $u\phi$ also defines a self-map of $M$ covering $(t_{6P})^{-1}$ as in Definition 2.3. Since the operator norm of $u\phi_x$ is strictly less than 1, the action is contracting with respect to the Hermitian norm function $M^{\times,\circ} \to \mathbb{R}_{>0}$. The $\mathbb{Z}$-action on this open set is therefore properly discontinuous. □

**Definition 2.5.** Let $\phi_0$ be a linearization as in Corollary 2.2 and let $\phi = u\phi_0$ with $|u| < c$, $u \in \mathbb{C}^*$ as in Proposition 2.4. We define a smooth threefold as the quotient

$$X^\circ(u) := M^{\times,\circ}/\mathbb{Z}$$

of the $\mathbb{Z}$-action of Definition 2.3 associated to the linearization $\phi$.

**Proposition 2.6.** *$X^\circ(u) \to \mathbb{A}^1$ is a fibration with smooth complex 2-torus fibers over $\mathbb{A}^1 \setminus \{0,1\}$.*

*Proof.* Over $t \in \mathbb{A}^1 \setminus \{0,1\}$ we have an exact sequence

$$1 \to \mathbb{C}^* \to M_t^\times \to S_t \to 0, \tag{2.1}$$

i.e. the fiber of $M^\times$ over $t$ is a semiabelian variety. So the universal cover of $M_t^\times$ is $\mathbb{C}^2$, and its fundamental group is isomorphic to $\mathbb{Z}^3$.

Quotienting by the additional $\mathbb{Z}$-action associated to the linearization $\phi$ gives a complex 2-torus, because the action is properly discontinuous, see Proposition 2.4, and the additional $\mathbb{Z}$-factor increases the rank of the subgroup of $\mathbb{C}^2$ from 3 to 4. □

**Proposition 2.7.** *In a punctured neighborhood of $\infty \in \mathbb{P}^1$ with coordinate $q$, $X^\circ(u)_q$ is biholomorphic to*

$$(\mathbb{C}^*)^2/\langle (c_1q, c_2), (c_3, c_4q)\rangle$$

*for holomorphic units $c_i \in \mathbb{C}^* + O(q)$ on $\Delta$.*

*Proof.* Since $S_\infty$ is an $I_1$ Kodaira fiber, the restriction of $S$ to $\Delta^*$ admits a Tate uniformization $S_q \simeq \mathbb{C}^*/q^{\mathbb{Z}}$. After pulling $M^\times$ back to the partial cover $\mathbb{C}^* \to S_q$ and choosing a trivialization, its total space is $(\mathbb{C}^*)^2$. The deck transformation defining the Tate curve acts by

$$(z,w) \mapsto (c_1 qz, c_2 w), \text{ for } (z,w) \in (\mathbb{C}^*)^2.$$

Here $c_1$ and $c_2$ are holomorphic units. The first scaling factor has order one in $q$ because it is the Tate period, while the second has order zero because $M$ has degree zero and extends across the $I_1$ fiber.

The linearization $\phi$ gives a second transformation of $(\mathbb{C}^*)^2$. It covers translation by $-6P$ on the Tate curve. Since $6P$ specializes to a point of the smooth locus of the $I_1$ fiber, its multiplicative Tate coordinate is a holomorphic unit. Thus the first component of this transformation has the form $z \mapsto c_3 z$. Moreover, $\phi$ vanishes to first order along $S_\infty$ so its action in the fiber direction of $M^\times$ has the form $w \mapsto c_4 qw$, where $c_4$ is a holomorphic unit. Hence the second transformation is

$$(z,w) \mapsto (c_3 z, c_4 qw), \text{ for } (z,w) \in (\mathbb{C}^*)^2.$$

Quotienting $(\mathbb{C}^*)^2$ by these two commuting transformations gives the asserted description of $X^\circ(u)_q$ for $q \in \Delta^*$. $\square$

**Proposition 2.8.** *A $\mathbb{Z}^2$-periodic tiling of $\mathbb{R}^2$ by lattice simplices of area $\frac{1}{2}$ determines a Kulikov ($K$-trivial, semistable) filling*

$$X^\circ(u) \hookrightarrow X(u)$$

*over $\infty \in \mathbb{P}^1$. For example, one may take the tiling by triangles in Figure 2. In this case, the fiber $X(u)_\infty$ is the result of gluing the opposite sides of a hexagon of $(-1)$-curves on a del Pezzo surface $dP_6$.*

*Proof.* This follows from a simple Mumford fan construction [Mum72]. By Proposition 2.7, the fiber over the punctured disk is $(\mathbb{C}^*)^2$ modulo the subgroup generated by the rows of the matrix

$$c \cdot q^{\left(\begin{smallmatrix}1 & 0\\ 0 & 1\end{smallmatrix}\right)}, \qquad c = \begin{pmatrix} c_1 & c_2 \\ c_3 & c_4 \end{pmatrix},$$

where $\cdot$ denotes entrywise multiplication. Thus, a $\mathbb{Z}(1,0) \oplus \mathbb{Z}(0,1) \simeq \mathbb{Z}^2$-periodic tiling of $\mathbb{R}^2$ provides a Mumford construction filling the family over $q = 0$. Take the triangulation of the unit square into two triangles. Extending $\mathbb{Z}^2$-periodically gives such a tiling $\mathcal{T}$. Put $\mathcal{T}$ at height 1 in $\mathbb{R}^3$, i.e. $\mathbb{R}^2 \times \{1\} \subset \mathbb{R}^2 \times \mathbb{R} \simeq \mathbb{R}^3$, and take the cone $\mathrm{Cone}(\mathcal{T})$. The result is a $\mathbb{Z}^2$-periodic fan in $\mathbb{R}^3$ with standard affine cones.

This fan defines a one-parameter Mumford construction of degenerating complex 2-tori, as in [EGFS25, Constr. 3.3 and Rem. 3.5]: we take the $\mathbb{Z}^2$-quotient of the toric variety of $\mathrm{Cone}(\mathcal{T})$, for which the

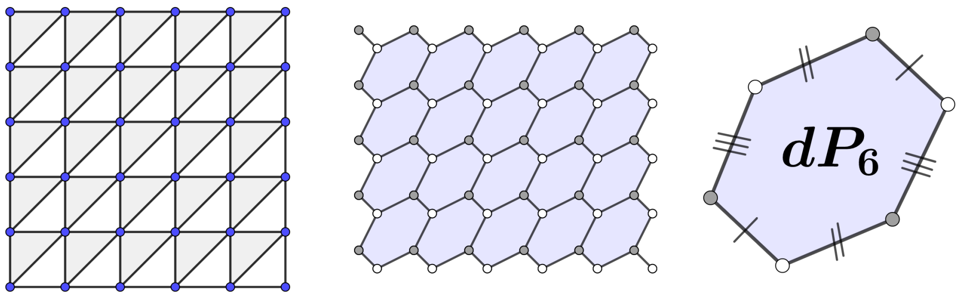


FIGURE 2. Left: $\mathbb{Z}^2$-periodic tiling $\mathcal{T}$ of the plane $\mathbb{R}^2$. Middle: $\mathbb{Z}^2$-periodic infinite quilt of smooth toric surfaces. Right: Its $\mathbb{Z}^2$-quotient and the central fiber of Mumford construction, a glued $dP_6$. Two 0-dimensional torus orbits in white and grey, three 1-dimensional torus orbits in black; the 2-dimensional torus orbit in blue.

fiber $(\mathbb{C}^*)^2$ over $q \in \Delta^*$ of the natural map to $\mathbb{A}^1$ is quotiented by the $\mathbb{Z}^2$-action of Proposition 2.7.

Before quotienting, the central fiber is an infinite periodic quilt of smooth toric surfaces, whose components are isomorphic to $dP_6$ (see Figure 2). The passage to the $\mathbb{Z}^2$-quotient has the effect of taking a single such $dP_6$ and gluing the opposite sides of its anticanonical hexagon, as described in the proposition. $\square$

**Proposition 2.9.** *The fibration $X(u) \to \mathbb{P}^1$ admits a holomorphic section. Over $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, this section may be taken as the origin of the smooth complex 2-torus fibers.*

*Proof.* The restriction of $M$ to the zero section $O$ admits a section $e$ which is non-vanishing over $\mathbb{A}^1$ and has a first order zero at $\infty$, because $M \cdot O = 1$. Hence $e$ defines a section of $M|_O \to O$, valued in $M^\times|_O$ away from $\infty$. Its image under the quotient map of Definition 2.5 defines a section of $X^\circ(u) = M^{\times,\circ}/\mathbb{Z} \to \mathbb{A}^1$.

Near $\infty$, the condition that $e$ vanishes to first order implies that it defines a local section of the uniformization of Proposition 2.7. Indeed, we may first translate $e$ by the period $(c_3^{-1}, c_4^{-1}q^{-1})$. This translated section has nonzero limit at $q = 0$, and so extends as a section of the Mumford construction. It coincides with $e$ in the quotient $X^\circ(u)$. $\square$

We have thus completed the summary of the "initial construction": We have produced a smooth, compact complex threefold $X(u)$ for all sufficiently small $u \in \mathbb{C}^*$, which is a complex 2-torus fibration, with section, and three singular fibers.

## 3. Logarithmic modification at 0 and 1

We now fill the family $X(u) \to \mathbb{P}^1$ at 0 and 1 in a new way. This logarithmic transform changes the fiber without changing the torus family over the punctured neighborhoods of 0 and 1, see e.g. [BHPV04, Ch. V, §13], [EFGMS25, §5.3].

To this end, we first describe the family $X(u)$ over a neighborhood of 0 and 1. Observe that the $\mathrm{SL}_2(\mathbb{Z})$ monodromies of $S \to \mathbb{P}^1$ about 0 and 1 have orders 3 and 4, respectively.

**Proposition 3.1.** *Let $\Delta_0 \ni 0$, $\Delta_1 \ni 1$ be disks. Consider the order 3, resp. 4 base change $\Delta \to \Delta_0$, $s \mapsto s^3 = t$ or $\Delta \to \Delta_1$, $s \mapsto s^4 = t$.*

(1) *$S \times_{\Delta_i} \Delta$ admits a smooth elliptic reduction $S'$ for $i = 0, 1$ isomorphic over the origin of $\Delta$ to, respectively, the triangular or square elliptic curve $E_\triangle$ or $E_\square$.*

(2) *The linearization, viewed as a biholomorphism $\phi \colon M^{\times,\circ} \to M^{\times,\circ}$, lifts to a biholomorphism $\phi'$ of $M' := \mathcal{O}_{S'}(P' - O')$ where $P'$ and $O'$ are the transforms of $P$ and $O$.*

*Furthermore, the action of $\phi'$ on $E_\triangle$ or $E_\square$ is trivial, and $\phi'$ acts on the $\mathbb{G}_m$-torsor $M'^{\times}$ (which over $E_\triangle$ or $E_\square$ corresponds to either a 3-torsion or a 2-torsion line bundle $L^\times$, respectively) by a scalar in $\mathbb{C}^*$.*

*Proof.* The smooth reduction of $IV^*$ and $III$ Kodaira fibers after a monodromy-trivializing base change is a well-known base change computation, see [BHPV04, Ch. V, §10]. Let $S''$ be a resolution of indeterminacy of $S \times_{\Delta_i} \Delta \dashrightarrow S'$. The biholomorphism $\phi$ pulls back to a biholomorphism of the $\mathbb{G}_m$-torsor $M''^{\times}$ over $S''$ corresponding to the pullback of $M$. Since $t_{6P}$ defines an element of $\mathrm{Aut}^0$ of every fiber, the lift of $\phi$ to a biholomorphism of $M''^{\times}$ preserves the components contracted over the morphism $S'' \to S'$ (on which $M''$ is trivial).

Hence this lift descends to a linearization $\phi'$ of $M'^{\times}$, which is equivariant over the action of $t_{6P'}$ on $S'$. By an explicit computation of $S''$ and the smooth model $S'$, it is easily seen that $P'$ intersects a 3-torsion point of $E_\triangle$ and a 2-torsion point of $E_\square$. Hence $t_{6P'}$ acts trivially on the central fiber $E_\triangle$ or $E_\square$ and the line bundle $M'|_{E_\triangle \text{ or } E_\square}$ is either a 3-torsion or 2-torsion element $L \in \mathrm{Pic}^0(E_\triangle \text{ or } E_\square)$ of the Picard group.

It follows that on the central fiber, $\phi'$ is simply an automorphism of the $\mathbb{G}_m$-torsor $L^\times$ (acting trivially on the base). □

**Corollary 3.2.** *The quotient $L^\times/\mathbb{Z}$ arising from Proposition 3.1 is an abelian surface $(\widetilde{E}_\triangle \times E_0')/\mu_3$ or $(\widetilde{E}_\square \times E_1')/\mu_2$.*

*Proof.* As $L$ is a 3- or 2-torsion bundle over $E_\triangle$ or $E_\square$, its pullback to an étale $\mu_3$- or $\mu_2$-cover

$$\widetilde{E}_\triangle \to E_\triangle \text{ or } \widetilde{E}_\square \to E_\square$$

is trivial. Along this pullback, the $\mathbb{Z}$-action lifts to an action on $\mathbb{C}^*$, whose quotient is an elliptic curve $E_i'$. The deck group of the étale cover acts by translation on both the vertical and horizontal factors. So the quotient is an abelian surface (rather than bielliptic). $\square$

Abstractly, $\widetilde{E}_\triangle \simeq E_\triangle$ and $\widetilde{E}_\square \simeq E_\square$. We retain the tildes because the covering maps are nontrivial isogenies of degrees 3 and 2, respectively.

**Corollary 3.3.** *The order* 3*, resp.* 4*, base change of* $X(u) \to \mathbb{P}^1$ *locally near* 0*, resp.* 1*, admits a bimeromorphic modification* $X' \to \Delta$ *to a smooth complex* 2*-torus fibration, with filling as in Corollary 3.2.*

The deck group $\mu_3$, resp. $\mu_4$, acts on the smooth reduction $S' \to \Delta$, and we may recover $S|_{\Delta_i}$ as a relatively minimal model of the quotient. The same holds for $X(u)$ and the filling of Corollary 3.3.

Let $\Lambda \coloneqq H_1(X(u)_t, \mathbb{Z})$ be a smooth complex 2-torus fiber, for a base point $t \in \mathbb{P}^1 \backslash \{0, 1, \infty\}$. Let $\gamma_0, \gamma_1, \gamma_\infty \in \pi_1(\mathbb{P}^1 \backslash \{0, 1, \infty\}, t)$ be standard counterclockwise based meridians, ordered so that $\gamma_0\gamma_1\gamma_\infty = 1$. Write the monodromy representation as

$$T_i \coloneqq \mathrm{Mon}(\gamma_i) \in \mathrm{SL}(\Lambda) \simeq \mathrm{SL}_4(\mathbb{Z}).$$

Let us comment a bit more on the structure of $\Lambda$ and the rank 4, weight $-1$ VHS on the corresponding local system. By construction, the complex 2-tori over a general point of $\mathbb{P}^1$ are $\mathbb{Z}$-quotients of the semiabelian variety $M_t^\times$ which fit into the exact sequence (2.1). Thus we have a natural filtration:

$$0 \to \Lambda^\times \to \Lambda \xrightarrow{\psi} \mathbb{Z} \to 0 \tag{3.1}$$

where $\Lambda^\times = \pi_1(M_t^\times)$ and a further exact sequence

$$0 \to \mathbb{Z} \to \Lambda^\times \to \Lambda' \to 0 \tag{3.2}$$

where $\Lambda'$ is the rank 2, weight $-1$ local system underlying the $\mathbb{Z}$-VHS of the elliptic fibration $S \to \mathbb{P}^1$. After good reduction at 0 or 1, it specializes to $H_1(E_\triangle, \mathbb{Z})$ or $H_1(E_\square, \mathbb{Z})$.

**Proposition 3.4.** *We have conjugacies*

$$T_0 \sim_{\mathbb{Q}} \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \; T_1 \sim_{\mathbb{Q}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \; T_\infty \sim_{\mathbb{Z}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

*Proof.* After the cyclic base changes of orders 3 and 4, the family has good reduction at 0 and 1. Let $F_0$ and $F_1$ denote the resulting smooth central fibers. The isogenies in Corollary 3.2 give decompositions

$$\Lambda \simeq H_1(F_i, \mathbb{Z}) \supset H_1(\widetilde{E}_{\triangle \text{ or } \square}, \mathbb{Z}) \oplus H_1(E_i', \mathbb{Z}). \tag{3.3}$$

Put $\rho = e^{2\pi i/3}$. With the complex orientations, the deck transformation acts on the first summand by multiplication by $\rho$ in the basis $(1, \rho)$ at 0 (type $IV^*$), and by multiplication by $-i$ in the basis $(1, i)$ at 1 (type $III$). It acts trivially on the second summand. Hence

$$T_0 \sim_{\mathbb{Q}} \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ and } T_1 \sim_{\mathbb{Q}} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with the $\mathbb{Z}$-conjugacy class determined in Section 5 from the explicit overlattice $\Lambda$. At $\infty$, Proposition 2.7 identifies the family with the quotient of $(\mathbb{C}^*)^2$ by the Tate period and the linearization period, having respective $q$-orders $(1,0)$ and $(0,1)$. So

$$T_\infty \sim_{\mathbb{Z}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We determine the monodromy representation exactly in Section 5. $\square$

Denote the *local monodromy invariants* by

$$\Lambda^{T_i} = \{\lambda \in \Lambda \,:\, T_i(\lambda) = \lambda\}.$$

The finite group $\frac{1}{3}\Lambda^{T_0}/\Lambda^{T_0}$ consists of the 3-torsion translations on the base change $X(u) \times_{\Delta_0} \Delta$, which are invariant under the $\mu_3$ deck action, while the finite group $\frac{1}{4}\Lambda^{T_1}/\Lambda^{T_1}$ consists of the 4-torsion translations on the base change $X(u) \times_{\Delta_1} \Delta$, which are invariant under the $\mu_4$ deck action. These groups of torsion sections play an important role in the following construction.

**Construction 3.5** (Log transform)**.** Given $\overline{a}_0 \in \frac{1}{3}\Lambda^{T_0}/\Lambda^{T_0}$ we may define a twisted $\mu_3$-action $\widetilde{\rho}$ on the smooth reduction $X' \to \Delta$ of the base change $X(u) \times_{\Delta_0} \Delta$ by setting

$$\widetilde{\rho}(x) = \rho(x) + \overline{a}_0.$$

Here $\rho$ is the generator of the original deck action of $\mu_3$ coming from the base change $\Delta \to \Delta_0$. We may then form the twisted quotient

$$X'/\langle \widetilde{\rho} \rangle.$$

We require the following lemma:

**Lemma 3.6.** *The twisted quotient $X'/\langle\widetilde{\rho}\rangle$ is biholomorphic over the punctured neighborhood $\Delta_0^*$ to $X(u)|_{\Delta_0^*}$. Furthermore, a lift of $\overline{a}_0$ to some $a_0 \in \frac{1}{3}\Lambda^{T_0}$ determines explicitly such a biholomorphism.*

*Proof.* Put $v_0 = 3a_0 \in \Lambda^{T_0}$ and choose the generator of the deck group so that it acts on the base by $s \mapsto \zeta_3 s$. Over $\Delta^*$ define the section

$$\sigma_{a_0}(s) = \frac{\log s}{2\pi i}v_0 = \frac{3\log s}{2\pi i}a_0.$$

Although $\log s$ is multivalued, moving to a new branch changes $\sigma_{a_0}$ by an integral multiple of $v_0 \in \Lambda$. It therefore defines a single-valued holomorphic section of the family of complex tori over $\Delta^*$. Let

$$\Psi_{a_0}(x,s) = (x + \sigma_{a_0}(s), s)$$

be translation by this section.

Since $v_0$ is fixed by $T_0$, analytic continuation gives

$$\sigma_{a_0}(\zeta_3 s) = T_0\sigma_{a_0}(s) + a_0.$$

Consequently,

$$\Psi_{a_0} \circ \rho = \widetilde{\rho} \circ \Psi_{a_0}.$$

Thus $\Psi_{a_0}$ descends to a biholomorphism between the quotients by $\rho$ and $\widetilde{\rho}$ over $\Delta_0^*$. The quotient by $\rho$ is $X(u)|_{\Delta_0^*}$ which proves the lemma. □

There is an exactly analogous construction given an element $\overline{a}_1 \in \frac{1}{4}\Lambda^{T_1}/\Lambda^{T_1}$ and a lift $a_1 \in \frac{1}{4}\Lambda^{T_1}$. The $a_i$ encode the "multiplicity class" of [EFGMS25, Def. 5.22].

**Definition 3.7.** We define the analytic space $X(u, a_0, a_1)$ by gluing to $X(u)|_{\mathbb{P}^1\setminus\{0,1\}}$ the twisted quotients of Lemma 3.6 associated to $a_0 \in \frac{1}{3}\Lambda^{T_0}$ and $a_1 \in \frac{1}{4}\Lambda^{T_1}$.

**Remark 3.8.** By construction, $X(u, 0, 0)$ is bimeromorphic to $X(u)$.

**Corollary 3.9.** *When $\overline{a}_0$ and $\overline{a}_1$ are primitive* 3-*, resp.* 4-*torsion, the twisted action of $\widetilde{\rho}$ on $X_i'$, for $i = 0, 1$, is free. Thus, the fiber of*

$$X(u, a_0, a_1) \to \mathbb{P}^1$$

*is multiple, of*

(1) *multiplicity* 3 *over* 0*, with reduced fiber a bielliptic surface of Bagnera-de Franchis type $\mu_3 \times \mu_3$.*
(2) *multiplicity* 4 *over* 1*, with reduced fiber a bielliptic surface of Bagnera-de Franchis type $\mu_2 \times \mu_4$.*

See [BHPV04, Ch. V, §5] for the classification of bielliptic surfaces.

*Proof.* See Construction 3.5 and Corollary 3.2. □

**Lemma 3.10.** *The space of globally monodromy-invariant linear functionals $\Lambda \to \mathbb{Z}$ has rank 1, generated by the map $\psi\colon \Lambda \to \mathbb{Z}$ of (3.1).*

*Proof.* This is immediate from the monodromy matrices $[T_0]_{\mathcal{B}_0}$, $[T_1]_{\mathcal{B}_1}$, and change of basis $C$ computed in Theorem 5.1. □

We will see the significance of this global invariant covector in the course of the proof of the main theorem.

## 4. Main result

**Theorem 4.1.** *Let $Y \coloneqq X(u, a_0, a_1)$ where local monodromy invariants $a_0 \in \frac{1}{3}\Lambda^{T_0}$ and $a_1 \in \frac{1}{4}\Lambda^{T_1}$ are chosen so that:*

(1) *$\overline{a}_0$ and $\overline{a}_1$ are, respectively, primitive 3- and 4-torsion,*
(2) $|12\psi(a_0 + a_1)| = 1$.

*Then for all $u \in \Delta^*$ in a sufficiently small punctured disk,*

$$Y \simeq_{\text{diffeo}} \mathbb{S}^6$$

*is a compact complex manifold, diffeomorphic to the 6-sphere.*

Conditions (1) and (2) can in fact be satisfied. For instance, we may take $a_0 = \frac{1}{3}e_4$ and $a_1 = -\frac{1}{4}f_4$ in the notation of Theorem 5.1.

*Proof.* By Section 6, $Y$ is simply connected. By Section 7, it has the integral homology of $\mathbb{S}^6$. The Hurewicz theorem implies that $Y$ is a homotopy 6-sphere. Smale's generalized Poincaré theorem identifies it topologically with $\mathbb{S}^6$ [Sma61].

By the $h$-cobordism theorem, oriented diffeomorphism classes of homotopy 6-spheres are classified by the group $\Theta_6$ [Mil65]. This group is trivial by the computation of Kervaire–Milnor [KM63, §7]. Hence $Y$ is also diffeomorphic to the standard 6-sphere. □

## 5. The monodromy representation

The overlattice $\Lambda$ in (3.3) is explicitly computable, with the containment in (3.3) of index 3 and 2 respectively. We now compute this overlattice, record integral normal forms of $T_0$ and $T_1$ in an appropriate oriented basis of $\Lambda$, and determine the change of basis between the bases expressing these two normal forms.

These computations are necessary for later steps, computing the fundamental group and integer homology of $Y$. Recall $\rho = e^{2\pi i/3}$.

**Theorem 5.1.** *Let*

$$e_1, e_2, \delta_0, e_4 = (1, 0), (\rho, 0), (0, 1), (0, \tau(E_0')) \in \mathbb{C}^2$$

*be a $\mathbb{Z}$-basis of $H_1(\widetilde{E}_\triangle, \mathbb{Z}) \oplus H_1(E'_0, \mathbb{Z})$, and let*

$$f_1, f_2, \delta_1, f_4 = (1,0), (i,0), (0,1), (0, \tau(E'_1)) \in \mathbb{C}^2$$

*be a $\mathbb{Z}$-basis of $H_1(\widetilde{E}_\square, \mathbb{Z}) \oplus H_1(E'_1, \mathbb{Z})$. The lattices of the two good reduction fibers are, respectively, the overlattices obtained by adjoining*

$$e_3 = \tfrac{1}{3}(2+\rho, 1), \qquad f_3 = \tfrac{1}{2}(1+i, 1).$$

*In the local bases $\mathcal{B}_0 = (e_1, e_2, e_3, e_4)$ and $\mathcal{B}_1 = (f_1, f_2, f_3, f_4)$ of $\Lambda$,*

$$[T_0]_{\mathcal{B}_0} = \begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad [T_1]_{\mathcal{B}_1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

*The circle classes of the principal $\mathbb{G}_m$-bundle $M^\times$ are*

$$\delta_0 = -2e_1 - e_2 + 3e_3, \qquad \delta_1 = -f_1 - f_2 + 2f_3.$$

*Furthermore, the columns of the matrix*

$$C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

*express $\mathcal{B}_1$ in the basis $\mathcal{B}_0$, and $C(\delta_1) = \delta_0$.*

*Proof.* The adjoining vectors $e_3$ and $f_3$ are determined by the requirement that their coordinates in $\widetilde{E}_\triangle$ or $\widetilde{E}_\square$ are non-zero and fixed by the order 3, resp. 4, automorphism, and that $\delta_0$ and $\delta_1$ represent the circle class of the principal $\mathbb{G}_m$-bundle $M^\times$.

Multiplication by $\rho$ sends $e_1 \mapsto e_2$ and $e_2 \mapsto -e_1 - e_2$, while fixing $\delta_0$ and $e_4$. Since $\delta_0 = 3e_3 - 2e_1 - e_2$, it follows that

$$T_0 e_1 = e_2, \qquad T_0 e_2 = -e_1 - e_2, \qquad T_0 e_3 = e_3 - e_1, \qquad T_0 e_4 = e_4.$$

This gives the displayed matrix for $T_0$. Similarly, multiplication by $-i$ sends $f_1 \mapsto -f_2$ and $f_2 \mapsto f_1$. Using $\delta_1 = 2f_3 - f_1 - f_2$, we obtain

$$T_1 f_1 = -f_2, \qquad T_1 f_2 = f_1, \qquad T_1 f_3 = f_3 - f_2, \qquad T_1 f_4 = f_4,$$

which gives the displayed matrix for $T_1$.

It remains to compare the two local markings in which we have the displayed normal forms. In $\Lambda' = \Lambda^\times / \mathbb{Z}\delta_i$, the monodromy representation of the underlying elliptic surface $S \to \mathbb{P}^1$ gives an equality

$$[f_1] = [e_1], \qquad [f_2] = \frac{[e_1] + 2[e_2]}{3}$$

in $\Lambda' \otimes \mathbb{Q}$ (rational coefficients arise because $[e_1], [e_2]$ and $[f_1], [f_2]$ generate sublattices of indices 3 and 2 in the homology lattices of the smooth reductions $E_\triangle$ and $E_\square$ over 0 and 1). Since $3[e_3] = 2[e_1] + [e_2]$ in $\Lambda'$, the second equality is equivalently

$$[f_1] = [e_1], \qquad [f_2] = [e_1 + e_2 - e_3]. \tag{5.1}$$

It remains to lift this identification through (3.2) and (3.1).

The restriction of $C$ to $\Lambda^\times$ is determined up to shears which act trivially on $\mathbb{Z}\delta_0$ and on $\Lambda'$. Thus there are integers $a, c$ such that

$$Cf_1 = e_1 + a\delta_0, \qquad Cf_2 = e_1 + e_2 - e_3 + c\delta_0.$$

The condition $C\delta_1 = \delta_0$, together with $\delta_1 = -f_1 - f_2 + 2f_3$, then gives

$$Cf_3 = e_3 + \tfrac{1}{2}(a + c)\delta_0.$$

Since $\delta_0$ is primitive, $a + c$ is even. Writing $a + c = 2b$, and then lifting from $\Lambda^\times$ to $\Lambda$ in a manner respecting (3.1), one obtains

$$\begin{aligned} Cf_1 &= e_1 + a\delta_0, \\ Cf_2 &= e_1 + e_2 - e_3 + (2b - a)\delta_0, \\ Cf_3 &= e_3 + b\delta_0, \\ Cf_4 &= e_4 + pe_1 + qe_2 + re_3 \end{aligned}$$

for some $a, b \in \mathbb{Z}$, with $p, q, r \in \mathbb{Z}$ determining the further lift from $\Lambda^\times$ to $\Lambda$. Equivalently,

$$C = \begin{pmatrix} 1 - 2a & 1 - 4b + 2a & -2b & p \\ -a & 1 - 2b + a & -b & q \\ 3a & -1 + 6b - 3a & 1 + 3b & r \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We now use the integral monodromy of the Mumford construction at $\infty$ to determine these five integers. Put

$$N_\infty = (T_0 C T_1 C^{-1})^{-1} - I.$$

The description in Proposition 3.4 of the monodromy of the Mumford degeneration implies that $N_\infty^2 = 0$ and that $\mathrm{im}(N_\infty)$ is a primitive rank 2 sublattice of $\Lambda$. Direct multiplication gives

$$N_\infty^2 e_j = a\delta_0 \quad (j = 1, 2, 3),$$

so $a = 0$. After setting $a = 0$, its remaining column is

$$N_\infty^2 e_4 = (3q + r)((1 - 4b)e_1 - 2be_2 + (6b - 1)e_3),$$

and hence $r = -3q$. Set $m = p - 2q$. The matrix of $N_\infty$ now becomes

$$\begin{pmatrix} 1 - 4b & 1 - 4b & 1 - 4b & 4bm \\ -2b & -2b & -2b & (2b - 1)m \\ 6b - 1 & 6b - 1 & 6b - 1 & (1 - 6b)m \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The greatest common divisor of its $2\times 2$ minors is $|m(1-6b)|$. Primitivity of $\mathrm{im}(N_\infty)$ therefore implies $b=0$ and $m=\pm 1$. Since $b=0$,

$$N_\infty e_j = e_1 - e_3 \quad (j=1,2,3), \qquad N_\infty e_4 = \pm(e_3 - e_2).$$

The first vanishing cycle comes from the $I_1$ monodromy of the elliptic surface, while the second comes from the linearization. For a counterclockwise meridian, the condition $\mathrm{ord}_\infty(\phi)=1$ gives $m=1$; a pole would give $m=-1$ and negate precisely the second contribution to the monodromy of the Mumford construction. It follows that

$$p = 2q+1, \qquad r = -3q,$$

and therefore

$$Cf_4 = e_4 + e_1 - q\delta_0.$$

The remaining integer $q$ is precisely the freedom to replace $e_4 \mapsto e_4 + k\delta_0$. This shear commutes with the local monodromy at 0, as both $e_4$ and $\delta_0$ are fixed. Thus, we may adjust the marking so that $q=0$. Then

$$\begin{aligned} Cf_1 &= e_1, & Cf_2 &= e_1 + e_2 - e_3,\\ Cf_3 &= e_3, & Cf_4 &= e_1 + e_4, \end{aligned}$$

which proves the displayed formula for $C$. □

Since $X(u) \to \mathbb{P}^1$ has a section, the two local monodromy matrices, together with the change of basis $C$, determine the topology of the real 4-torus bundle arising from restricting to $\mathbb{P}^1 \setminus \{0,1,\infty\}$.

## 6. The fundamental group

Put $U = \mathbb{P}^1 \setminus \{0,1,\infty\}$, and $J = Y|_U = X(u)|_U$. The section of Proposition 2.9 gives a base point in every fiber. Hence

$$\pi_1(J) = \Lambda \rtimes \pi_1(U),$$

where the action on $\Lambda$ is the monodromy representation. Write $\mathcal{B}_0 = (e_1,e_2,e_3,e_4)$ for the basis of Theorem 5.1, normalized so that

$$\psi(e_4) = 1, \qquad \psi(e_1) = \psi(e_2) = \psi(e_3) = 0.$$

We use the same letter $\psi$ for its extension to $\Lambda \otimes \mathbb{Q}$. In the common basis $\mathcal{B}_0$ we have

$$[T_1]_{\mathcal{B}_0} = C[T_1]_{\mathcal{B}_1}C^{-1} = \begin{pmatrix} -1 & 1 & -1 & 2\\ -1 & 0 & -1 & 1\\ 1 & 1 & 2 & -1\\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{6.1}$$

In particular, $\psi$ is fixed by both monodromies.

Since $T_\infty = (T_0T_1)^{-1}$, the preceding matrices give

$$(T_\infty - I)\Lambda = \langle e_1 - e_3, e_3 - e_2\rangle =: K_\infty.$$

These are precisely the two vanishing cycles (i.e. generators of $W_{-2}H_1$) of the Mumford construction at $\infty$. Thus $K_\infty \subset \pi_1(J)$ is killed by filling the fiber at $\infty$ (these cycles collapse to points at either of the two 0-dimensional toric strata of the glued $dP_6$). Since

$$T_0(e_1 - e_3) = e_1 + e_2 - e_3,$$

the three vectors $e_1 - e_3$, $e_3 - e_2$, and $e_1 + e_2 - e_3$ form a basis of $\ker\psi$. Hence the normal closure of $K_\infty$ is exactly

$$\ker\psi = \langle e_1, e_2, e_3\rangle.$$

Thus the image of $\Lambda$ in $\pi_1$ of the extension of $J$ over $\infty$ is infinite cyclic, generated by the image $c$ of $e_4$. Because $\psi$ is monodromy invariant, $c$ is central.

Put $v_0 = 3a_0 \in \Lambda^{T_0}$, $v_1 = 4a_1 \in \Lambda^{T_1}$, and set

$$m = \psi(v_0) = 3\psi(a_0) \in \mathbb{Z}, \qquad n = \psi(v_1) = 4\psi(a_1) \in \mathbb{Z}.$$

Let $x_0, x_1$ be the meridians $\gamma_0, \gamma_1$ about $0, 1$ lifted to the section of $J$. Since the section extends over the Mumford construction at $\infty$, we have the relation $x_0x_1 = 1$. At the multiple fibers, the two logarithmic fillings locally impose

$$x_0^3 = v_0, \qquad x_1^4 = v_1.$$

Including the neighborhood of the logarithmic filling into the total space $Y$, we in turn get the relation

$$x_0^3 = c^m, \qquad x_1^4 = c^n.$$

**Proposition 6.1.** *Let $p \coloneqq 12\psi(a_0 + a_1)$. The fundamental group of $Y$ is finite cyclic of order $|p|$, when $p \neq 0$. In particular, under the hypotheses of Theorem 4.1, $Y$ is simply connected.*

*Proof.* The preceding discussion gives

$$\pi_1(Y) = \left\langle c, x_0, x_1 \;\middle|\; c \text{ central}, \quad x_0x_1 = 1, \quad x_0^3 = c^m, \quad x_1^4 = c^n \right\rangle.$$

Eliminating $x_1 = x_0^{-1}$ makes the group abelian, generated by $x_0$ and $c$, and the remaining relations reduce this abelian group to a cyclic group of order $|4m + 3n|$. □

## 7. The integral homology

Let $f : Y \to \mathbb{P}^1$ be the torus fibration and put $V = \mathrm{Hom}(\Lambda, \mathbb{Z})$. Let $(e_1^*, e_2^*, e_3^*, e_4^*)$ be the basis of $V$ dual to $\mathcal{B}_0 = (e_1, e_2, e_3, e_4)$ (see Theorem 5.1). The filtration $\mathbb{Z}\delta_0 \subset \Lambda^\times \subset \Lambda$ induces the dual filtration

$$0 \subset W_0 \subset W_1 \subset W_2 = V, \qquad W_0 = (\Lambda^\times)^\perp, \qquad W_1 = (\mathbb{Z}\delta_0)^\perp.$$

Thus $W_0 = \mathbb{Z}e_4^* = \mathbb{Z}\psi$ and, since $\delta_0 = -2e_1 - e_2 + 3e_3$,

$$W_1 = \langle e_1^* - 2e_2^*,\ 3e_2^* + e_3^*,\ e_4^* \rangle .$$

Over $U$, the local system $R^k f_* \mathbb{Z}$ has fiber $\wedge^k V$. Recall that the fibers over 0, 1 are multiple with bielliptic reductions, while the fiber at $\infty$ is the Mumford construction central fiber of Section 2.

**Proposition 7.1.** $\chi_{\mathrm{top}}(Y) = 2$.

*Proof.* Since a complex 2-torus and both reduced bielliptic fibers have Euler characteristic zero, additivity of the Euler characteristic reduces the computation to the central fiber $Y_\infty$ of the Mumford construction. Its toric stratification has exactly two zero-dimensional strata. Positive-dimensional strata contain a factor $\mathbb{C}^*$ and hence have Euler characteristic zero. So $\chi_{\mathrm{top}}(Y) = \chi_{\mathrm{top}}(Y_\infty) = 2$. □

**Lemma 7.2.** *The monodromy preserves the primitive alternating form*

$$\xi = 3e_1^* \wedge e_2^* + e_1^* \wedge e_3^* - 2e_2^* \wedge e_3^* - 2e_3^* \wedge e_4^*.$$

*Its elementary divisors are* $(1, 6)$.

*Proof.* Taking inverse transposes of the matrices of Theorem 5.1 shows directly that $T_i^* \xi = \xi$ for $i = 0, 1, \infty$. Computing $\frac{1}{2}\xi \wedge \xi = -6e_1^* \wedge e_2^* \wedge e_3^* \wedge e_4^*$ proves that the elementary divisors are $(1, 6)$. □

**Lemma 7.3.** *The monodromy-invariant classes over $U$ are*

$$\begin{aligned} H^0(U, R^1 f_* \mathbb{Z}) &= \mathbb{Z}\psi \quad (= W_0), \\ H^0(U, R^2 f_* \mathbb{Z}) &= \mathbb{Z}\xi, \\ H^0(U, R^3 f_* \mathbb{Z}) &= \mathbb{Z}\theta \quad (= \wedge^3 W_1), \end{aligned}$$

*where* $\theta \coloneqq \xi \wedge \psi$.

*Proof.* The first equality is Lemma 3.10. The invariance of $\xi$ is Lemma 7.2, and the invariance of $\theta = \xi \wedge \psi$ follows. Note that $\wedge^3 W_1$ is also monodromy invariant by general principle. Performing a linear algebra check with the matrices $[T_0]_{\mathcal{B}_0}$ and $[T_1]_{\mathcal{B}_0}$ of Theorem 5.1 and (6.1), one can also directly compute that the classes $\psi$, $\xi$, $\theta$ integrally generate the monodromy invariants in $\wedge^k V$, for $k = 1, 2, 3$, respectively. □

We next determine what multiples of these invariant classes extend over the three singular fibers.

**Lemma 7.4.** *Suppose $|p| = 1$. For $i = 0, 1$, let $\pi_i : F_i \to G_i$ be the unramified cover from the smooth good reduction fiber to the reduced bielliptic fiber. Write*

$$\mathrm{sp}_i^k \coloneqq \pi_i^* : H^k(G_i, \mathbb{Z}) \to H^k(F_i, \mathbb{Z})^{T_i} = (\wedge^k V)^{T_i}.$$

*Then*

$$[(\wedge^k V)^{T_i} : \mathrm{im}(\mathrm{sp}_i^k)] = \begin{array}{c|ccc} i\backslash k & 1 & 2 & 3 \\ \hline 0 & 3 & 1 & 1 \\ 1 & 4 & 2 & 2 \end{array} \tag{7.1}$$

*with cokernel generated by the image of $\psi$, $\xi$, $\theta$ in degrees $k = 1, 2, 3$. At $\infty$, the specialization image is the full $T_\infty$-invariant lattice $(\wedge^k V)^{T_\infty}$ in all three degrees $k = 1, 2, 3$.*

*Proof.* The local invariant lattices are

$$\begin{aligned} V^{T_0} &= \langle e_3^*, e_4^* \rangle, & (\wedge^2 V)^{T_0} &= \langle e_3^* \wedge e_4^*, \xi \rangle, \\ V^{T_1} &= \langle e_2^* + e_3^*, e_4^* \rangle, & (\wedge^2 V)^{T_1} &= \langle (e_2^* + e_3^*) \wedge e_4^*, \xi \rangle. \end{aligned} \tag{7.2}$$

Let $d_0 = 3$ and $d_1 = 4$. The affine group $\pi_1(G_i)$ is generated by the translations $t_\lambda$ and by $\widetilde{\rho}_i$ (see Construction 3.5), with relations

$$\widetilde{\rho}_i t_\lambda \widetilde{\rho}_i^{-1} = t_{T_i \lambda}, \qquad \widetilde{\rho}_i^{d_i} = t_{v_i}$$

where, recall, $v_i = d_i a_i$ for $i = 0, 1$. Thus

$$\mathrm{im}(\mathrm{sp}_i^1) = \{\alpha \in V^{T_i} : d_i \mid \alpha(v_i)\}.$$

Since $|p| = 1$, $e_4^*(v_0) = m$ is a unit modulo 3 and $e_4^*(v_1) = n$ is a unit modulo 4. Evaluation on $v_i$ therefore gives

$$V^{T_i} / \mathrm{im}(\mathrm{sp}_i^1) \simeq \mathbb{Z}/d_i\mathbb{Z},$$

generated by the class of $e_4^*$. The first column of (7.1) follows.

The indices in degree 2 follow from computing the intersection pairings on $H^2(F_i, \mathbb{Z})$ and $H^2(G_i, \mathbb{Z})$. The Gram matrices of the intersection pairing of the bases (7.2) of $H^2(F_i, \mathbb{Z})^{T_i}$ are

$$\begin{pmatrix} 0 & 3 \\ 3 & -12 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 2 \\ 2 & -12 \end{pmatrix}.$$

The Bagnera–de Franchis types of $G_i$ ensure that $H^2(G_i, \mathbb{Z})$ is free of rank 2 [Ser90, Rem. 1.6]. By Poincaré duality, the intersection pairing on its middle cohomology is unimodular.

Write $(\pi_i)_*$ for proper pushforward. It satisfies $(\pi_i)_*\pi_i^* = d_i$ and so every $\mathrm{sp}_i^k$ is injective. Since pullback multiplies the intersection pairing by $d_i$, comparison of determinants gives

$$[(\wedge^2 V)^{T_0} : \mathrm{im}(\mathrm{sp}_0^2)] = 1, \qquad [(\wedge^2 V)^{T_1} : \mathrm{im}(\mathrm{sp}_1^2)] = 2.$$

The latter cokernel is generated by $\xi$: if $\xi$ descended at the order 4 fiber, its square downstairs would be $\xi^2/4 = -3$, but the intersection form of $G_1$ is even (as $K_{G_1} \sim_{\mathbb{Q}} 0$). The second column of (7.1) follows.

Finally, we make the cohomology degree 3 computation. The local invariant lattices in degree 3 are

$$(\wedge^3 V)^{T_0} = \langle e_1^* \wedge e_2^* \wedge e_3^*,\ \theta\rangle\,,$$
$$(\wedge^3 V)^{T_1} = \langle e_1^* \wedge e_2^* \wedge e_3^* - e_2^* \wedge e_3^* \wedge e_4^*,\ \theta\rangle\,.$$

Pairing these bases with the respective bases of $V^{T_i}$ in (7.2) gives

$$\begin{pmatrix} 0 & 3 \\ -1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 2 \\ -1 & 0 \end{pmatrix}.$$

Let $b_i = [(\wedge^3 V)^{T_i} : \mathrm{im}(\mathrm{sp}_i^3)]$. The pairing between $H^1(G_i, \mathbb{Z})$ and $H^3(G_i, \mathbb{Z})$ is unimodular, whereas pullback to $F_i$ multiplies it by $d_i$. Since both groups have rank 2, the determinant of the pulled-back pairing is therefore $d_i^2$. On the other hand, replacing the two invariant lattices by sublattices of indices $d_i$ and $b_i$ multiplies the determinant by $d_i b_i$. The two displayed pairing matrices therefore give

$$3 \cdot b_0 \cdot 3 = 3^2, \qquad 4 \cdot b_1 \cdot 2 = 4^2.$$

Thus $b_0 = 1$ and $b_1 = 2$. Finally, the description of $\mathrm{im}(\mathrm{sp}_1^1)$ allows us to choose $\ell \in \mathbb{Z}$ so that $e_2^* + e_3^* - \ell e_4^*$ descends to $G_1$. Its pairing with $\theta$ is 2. If $\theta$ also descended, this pairing would be divisible by $d_1 = 4$, a contradiction. Hence the degree 3 cokernel at 1 is generated by the class of $\theta$. This proves the third column of (7.1).

At $\infty$, we have $\mathrm{im}\, N_\infty = \ker N_\infty = \ker(T_\infty - I)$, where $N_\infty := T_\infty - I$. Indeed, the $q$-order matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) \in \mathrm{GL}_2(\mathbb{Z})$ of Proposition 2.8 is unimodular. The last sentence of the lemma follows for $k = 1$. A computation of the Mayer–Vietoris spectral sequence for the toric stratification of $Y_\infty$ shows that the specialization maps to the monodromy invariant classes are also surjective for $k = 2, 3$. $\square$

We now take cohomology on the base.

**Lemma 7.5.** *Suppose* $|p| = 1$. *Then*

$$H^0(\mathbb{P}^1, R^1f_*\mathbb{Z}) = 12\mathbb{Z}\psi,$$
$$H^0(\mathbb{P}^1, R^2f_*\mathbb{Z}) = 2\mathbb{Z}\xi,$$
$$H^0(\mathbb{P}^1, R^3f_*\mathbb{Z}) = 2\mathbb{Z}\theta,$$

*and* $H^1(\mathbb{P}^1, R^1f_*\mathbb{Z}) = H^1(\mathbb{P}^1, R^2f_*\mathbb{Z}) = 0$. *If* $H^2(\mathbb{P}^1, \mathbb{Z}) = \mathbb{Z}\omega$, *then*

$$H^2(\mathbb{P}^1, R^0f_*\mathbb{Z}) = \mathbb{Z}\omega,$$
$$H^2(\mathbb{P}^1, R^1f_*\mathbb{Z}) = \mathbb{Z}\omega[e_1^* - e_2^*],$$
$$H^2(\mathbb{P}^1, R^2f_*\mathbb{Z}) = \mathbb{Z}\omega[(e_1^* - e_2^*) \wedge \psi].$$

*Proof.* Intersecting the three specialization lattices of Lemma 7.4 inside the invariant lattices of Lemma 7.3 gives the asserted $H^0$ groups.

For $j : U \hookrightarrow \mathbb{P}^1$ there is an exact sequence

$$\begin{gathered} 0 \to R^kf_*\mathbb{Z} \to j_*(\wedge^k V) \to Q^k \to 0, \text{ where} \\ Q^1 = (\mathbb{Z}/3\mathbb{Z})_0 \oplus (\mathbb{Z}/4\mathbb{Z})_1 \text{ and } Q^2 = Q^3 = (\mathbb{Z}/2\mathbb{Z})_1. \end{gathered} \tag{7.3}$$

Local-coefficient duality on the pair of pants gives

$$H^2(\mathbb{P}^1, j_*L) = L_{\langle T_0, T_1\rangle}$$

where the subscript denotes coinvariants. Row reduction gives

$$V_{\langle T_0,T_1\rangle} = \mathbb{Z}[e_1^* - e_2^*],$$
$$(\wedge^2 V)_{\langle T_0,T_1\rangle} = \mathbb{Z}[(e_1^* - e_2^*) \wedge \psi],$$
$$[\xi] = 12[(e_1^* - e_2^*) \wedge \psi].$$

Since the skyscraper sheaf $Q^k$ has no higher cohomology, this proves the assertions about $H^2$.

Finally we check the vanishing of the $H^1$ groups. We have a surjection $H^0(\mathbb{P}^1, j_*(\wedge^k V)) \to H^0(\mathbb{P}^1, Q^k)$, for $k = 1, 2$—the classes $\psi$, $\xi$ map to generators of $H^0(\mathbb{P}^1, Q^1) = \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, $H^0(\mathbb{P}^1, Q^2) = \mathbb{Z}/2\mathbb{Z}$ respectively by Lemma 7.4. From the long exact sequence of (7.3), it suffices to prove vanishing of $H^1(\mathbb{P}^1, j_*(\wedge^k V))$. For $L = V$ and $\wedge^2 V$, respectively, this group is identified with

$$\frac{(T_0 - I)L \cap (T_\infty - I)L}{(T_0 - I)(L^{T_1})}$$

if we normalize cocycles to vanish on $\gamma_1 \in \pi_1(U)$. For $L = V$ the numerator and denominator are $\mathbb{Z}(e_1^* + e_2^* + e_3^*)$; for $L = \wedge^2 V$ they are $\mathbb{Z}((e_1^* + e_2^* + e_3^*) \wedge \psi)$. Thus both groups vanish. $\square$

We now compute the differentials $d_2$ on the $E_2$-page of the Leray spectral sequence that contribute in total degree at most 3.

**Lemma 7.6.** *We have*

$$d_2(12\psi) = p\omega, \quad d_2(2\xi) \ = -p\omega[e_1^* - e_2^*], \quad d_2(2\theta) = p\omega[(e_1^* - e_2^*) \wedge \psi].$$

*Proof.* The first differential measures the incompatibility of local lifts of $12\psi$. The relations

$$\widetilde{\rho}_0^3 = t_{v_0}, \qquad \widetilde{\rho}_1^4 = t_{v_1}$$

show that the contributions from the two finite fibers are

$$\frac{12\psi(v_0)}{3} = 4m, \qquad \frac{12\psi(v_1)}{4} = 3n.$$

The fiber at $\infty$ contributes zero as $\psi$ extends. Hence, after fixing the common orientation, $d_2(12\psi) = p\omega$, where recall, $p = 4m + 3n$. Write

$$d_2(2\xi) = -p'\omega[e_1^* - e_2^*], \qquad d_2(2\theta) = p''\omega[(e_1^* - e_2^*) \wedge \psi].$$

The Leray spectral sequence is multiplicative and $d_2$ is a derivation. Hence, applying $d_2$ to

$$(2\xi)(12\psi) = 12(2\theta)$$

and using the identities

$$\xi \wedge \psi = \theta, \qquad [\xi] = 12[(e_1^* - e_2^*) \wedge \psi]$$

gives $p'' = 2p - p'$. If $\nu = e_1^* \wedge e_2^* \wedge e_3^* \wedge e_4^*$, then

$$(e_1^* - e_2^*) \wedge \theta = -\nu, \qquad \xi \wedge (e_1^* - e_2^*) \wedge \psi = -\nu.$$

Applying $d_2$ to $(2\xi)(2\theta) = 0$ therefore gives $2(p' - p'')\omega\nu = 0$. Hence $p' = p''$, and consequently $p' = p'' = p$. $\square$

**Theorem 7.7.** *If* $|12\psi(a_0 + a_1)| = 1$, *then*

$$H^k(Y, \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \leq k \leq 5. \end{cases}$$

*Thus* $Y$ *is an integral homology* 6*-sphere.*

*Proof.* The only differentials relevant in total degrees at most 3 are the three maps of Lemma 7.6. When $|p| = 1$, all three are isomorphisms. Lemma 7.5 then gives directly

$$H^1(Y, \mathbb{Z}) = H^2(Y, \mathbb{Z}) = H^3(Y, \mathbb{Z}) = 0.$$

Since $Y$ is closed and oriented of real dimension 6, Poincaré duality and universal coefficient theorem determine the remaining groups. $\square$

## 8. Additional remarks

**Remark 8.1.** Other than the discrete parameters in Theorem 4.1, there is the continuous parameter $u \in \Delta^*$. We may thus build a proper, holomorphic submersion $\mathcal{Y}^* \to \Delta^*$ whose fiber over $u$ is $X(u, a_0, a_1)$, and similarly, for the untwisted version, a proper holomorphic submersion $\mathcal{X}^* \to \Delta^*$ whose fiber over $u$ is $X(u)$.

For varying values of $u$, the complex manifold $X(u)$ is indeed deforming: Its algebraic dimension is 1 so the map to $\mathbb{P}^1$ is unique, rigidified by demanding that the singular fibers map over $0, 1, \infty$. Then, the periods of the fiber over $t \in \mathbb{P}^1 \setminus \{0, 1, \infty\}$ vary. Alternatively, the second elliptic curves $E_0'$ and $E_1'$ involved in the bielliptic fibers (Corollary 3.2) vary depending on the action of $\mathbb{Z}$ on $L^\times$ and hence depend on the scaling parameter $u\phi_0$ of the linearization, see Definition 2.5.

Does this family of complex manifolds degenerate in a nice way over the origin $0 \in \Delta$? The geometric construction is suggestive: The parameter $u$ is a rescaling parameter for the linearization $\phi\colon t_{6P}^*M \to M$ discussed in Section 2. Thus as we take the limit $|u| \to 0$, the modulus of the annular fundamental domain for multiplication by $u$ on $\mathbb{G}_m$ grows to infinity. Since, furthermore, the linearization $\phi$ lifts the translation $t_{6P}^{-1}$, this strongly suggests that the limit

$$X(0) \coloneqq \lim_{u \to 0} X(u)$$

can be constructed as follows: Take the $\mathbb{P}^1$-bundle $\mathbb{P}_S(\mathcal{O} \oplus M)$ and glue the zero section to the infinity section by $t_{6P}$. This can thus be viewed fiberwise over $t \in \mathbb{P}^1$ as a "compactified semiabelian" degeneration of the corresponding family of complex 2-tori $X_t(u)$ as $u \to 0$.

**Conjecture 8.2.** *There is a flat, proper filling $\mathcal{X} \to \Delta$ and the central fiber $X(0)$ is non-normal. The normalization is $\mathbb{P}_S(\mathcal{O} \oplus M)$ and the normalization map glues the zero and infinity sections via $t_{6P}$.*

For $\mathcal{Y}^* \to \Delta^*$, which is particularly appealing, given that it is a smooth $\mathbb{S}^6$-bundle over the punctured disk, it is less clear what the limit should be, but it is natural to guess that it is a logarithmic transform of $X(0)$, whose fibers over 0 and 1 have multiplicity 3 and 4, and whose reductions are "semiabelian type" degenerations of bielliptic surfaces.

**Remark 8.3.** The rank 4 local system of first homology groups of the fibers of $Y \to \mathbb{P}^1$ contains a natural rank 3 sub-local system, given by the kernel of the unique invariant (up to scale) covector $\psi$ of Lemma 3.10. Geometrically, this rank 3 sub-local system corresponds to a real 3-dimensional subtorus of the fibers. Quotienting fiberwise by the translational action of this subtorus, we reduce the fibers $Y_t$ to circles, but some care must be taken at $t = 0, 1, \infty \in \mathbb{P}^1$.

At $t = \infty$, the kernel of $\psi$ contains the vanishing cycles $W_{-2}H_1$ of the nearby fiber. The quotient by the vanishing cycles torus can be interpreted in the following manner: Take the Clemens collapse [Cle77] from the nearby fiber to $Y_\infty$ and then compose with the moment mapping on the glued $dP_6$ to a glued hexagon (which is a polytope of this Mumford construction).

There is, in the construction, a distinguished edge of the glued hexagon. Indeed, $\mathrm{Aut}^0(Y) \simeq \mathbb{C}^*$ (acting by scaling on the prequotient $M^\times$ as a $\mathbb{G}_m$-torsor) and the fixed locus of this $\mathbb{C}^*$-action is one of the three glued edges. This distinguished edge determines a further quotient of the hexagonal torus to a circle. Thus over $\infty$, this quotient map of the first paragraph extends continuously to the composition of the moment mapping with a further quotient to a circle.

At 0 or 1, the situation is different. Here, the rank 3 sub-local system contains the local rank 2 sub-local system on which the monodromy about 0 or 1 is nontrivial. Quotienting by the corresponding subtorus gives a multiple elliptic curve $E'/\mu_3$ or $E'/\mu_4$ because the twisted action defining the logarithmic transform producing the bielliptic multiple fiber acts by a torsion translation on $E'$. The further quotient collapses $E'/\mu_3$ or $E'/\mu_4$ to a circle, with multiplicity 3 or 4.

The conclusion is that the quotient by the torus associated to this natural rank 3 sub-local system is a map $\mathbb{S}^6 \to Q$ where $Q$ is a Seifert fibered 3-manifold over $\mathbb{P}^1 \simeq \mathbb{S}^2$ with a multiple circle of multiplicity 3 over 0 and multiplicity 4 over 1. The condition $|12\psi(a_0 + a_1)| = 1$ in Theorem 4.1 translates exactly into the condition $Q \simeq \mathbb{S}^3$ with $\mathbb{S}^3 \to \mathbb{S}^2$ the Seifert fibration with multiplicities 3 and 4 over two points [Orl72, Ch. 1]. Thus the quotient map is a map $\mathbb{S}^6 \to \mathbb{S}^3$.

**Remark 8.4.** A very general smooth fiber $F$ of $X(u) \to \mathbb{P}^1$, or equivalently $Y \to \mathbb{P}^1$, is not an abelian variety, but does admit a monodromy invariant Hodge class

$$\xi \in H^{1,1}(F) \cap H^2(F, \mathbb{Z}).$$

The class $\xi$ (see Lemma 7.2) is thus the first Chern class of a line bundle. Its type, as a symplectic form, is $(1, 6)$. On $F^0H_1(F, \mathbb{C})$, the signature of the associated Hodge form $i\xi(v, \bar{v})$ is indefinite signature $(1, 1)$. Thus we have a natural period mapping

$$\mathbb{P}^1 \setminus \{0, 1, \infty\} \to \mathrm{Sp}(\mathbb{Z}^4, \xi)\backslash \mathrm{Sp}_4(\mathbb{R})/\mathrm{U}(1, 1).$$

The difference with the abelian case is that the action of $\mathrm{Sp}(\mathbb{Z}^4, \xi)$ on the domain of complex tori $\mathbb{D} \coloneqq \mathrm{Sp}_4(\mathbb{R})/\mathrm{U}(1,1)$ is not properly discontinuous. But, after all, the monodromy is not all of $\mathrm{Sp}(\mathbb{Z}^4, \xi)$—indeed, the monodromy preserves the filtration (3.1, 3.2) and so lies in a parabolic subgroup $P_{\mathbb{Z}} \subset \mathrm{Sp}(\mathbb{Z}^4, \xi)$. B. Klingler suggested the possibility that the monodromy group could act on $\mathbb{D}$ properly discontinuously (or at least does so, on an analytic open set containing the period image).

**Remark 8.5.** The construction in Section 2 (and 3) works in significantly greater generality than this one example. Indeed, consider an elliptic surface $\pi\colon S \to C$ with section, a line bundle $M \in \mathrm{Pic}(S)$ which lies in $\mathrm{Pic}^0$ of the general fiber, and a translation $t \in \mathrm{Aut}(S/C)$. Suppose $\mathcal{H}om(t^*M, M) \simeq \pi^*\mathcal{O}_C(\pm E)$ for $E$ an effective divisor (e.g. this holds when $C \simeq \mathbb{P}^1$ and $t$ preserves the components of the Kodaira fibers). We get an associated linearization $\phi \in \mathrm{Aut}(M^{\times,\circ})$ of the principal $\mathbb{G}_m$ bundle $M^\times$ over the complement of the fibers over $E$.

Then $\phi$ acquires either zeroes or poles along the inverse image of $E$, respectively. Scale $\phi$ by either a very small, or very large, complex number $u \in \mathbb{C}^*$, respectively. This action is properly discontinuous by the argument of Proposition 2.4 (since the linearization has only zeroes or only poles, some scaling is uniformly contracting or expanding). Furthermore, we can fill the quotient by a Mumford construction, as in Proposition 2.7. One may then perform log transforms, if desired. It is unclear what class of 6-manifolds can arise from such a construction.

A simple case of this is moving the zero of the linearization in the $\mathbb{S}^6$ construction above away from $\infty \in \mathbb{P}^1$. This has the effect, it appears, of contracting the $(-1,-1)$-curve in the Mumford fiber and then smoothing the resulting threefold ordinary double point. That is: $X(u)$ (and $Y$) admit a conifold transition.

## References

[ABGKR18] I. Agricola, G. Bazzoni, O. Goertsches, P. Konstantis, and S. Rollenske, *On the history of the Hopf problem*, Differential Geom. Appl. **57**, 1–9, 2018.

[AC26] Claude and L. Alpöge, *The* $(3,4,\infty)$ *modular family of* 2*-tori, completed at its three special points, is a complex structure on* $S^6$, available at https://alpo.ge/s6.pdf, 2026.

[BHPV04] W. Barth, K. Hulek, C. Peters, and A. Van de Ven, *Compact Complex Surfaces*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 4, Springer-Verlag, Berlin, 2004.

[BS53] A. Borel and J.-P. Serre, *Groupes de Lie et puissances réduites de Steenrod*, Amer. J. Math. **75**, 409–448, 1953.

[Bry21] R. L. Bryant, *S.-S. Chern's study of almost-complex structures on the six-sphere*, Internat. J. Math. **32**, no. 12, 2140006, 2021.

[CDP98] F. Campana, J.-P. Demailly, and T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compositio Math. **112**, 77–91, 1998.

[Cle77] C. H. Clemens, *Degeneration of Kähler manifolds*, Duke Math. J. **44**, 215–290, 1977.

[EFGMS25] P. Engel, S. Filipazzi, F. Greer, M. Mauri, and R. Svaldi, *Boundedness of some fibered $K$-trivial varieties*, arXiv:2507.00973, 2025.

[EGFS25] P. Engel, O. de Gaay Fortman, and S. Schreieder, *Combinatorics and Hodge theory of degenerations of abelian varieties: A survey of the Mumford construction*, arXiv:2507.15695, 2025.

[Gra97] A. Gray, *A property of a hypothetical complex structure on the six sphere*, Boll. Un. Mat. Ital. B (7) **11**, suppl. fasc. 2, 251–255, 1997.

[Hop48] H. Hopf, *Zur Topologie der komplexen Mannigfaltigkeiten*, in *Studies and Essays Presented to R. Courant on his 60th Birthday*, Interscience Publishers, New York, 167–185, 1948.

[KM63] M. A. Kervaire and J. W. Milnor, *Groups of homotopy spheres. I*, Ann. of Math. (2) **77**, 504–537, 1963.

[Mil65] J. Milnor, *Lectures on the $h$-Cobordism Theorem*, Princeton University Press, Princeton, 1965.

[Mir90] R. Miranda, *Persson's list of singular fibers for a rational elliptic surface*, Math. Z. **205**, 191–211, 1990.

[Mum72] D. Mumford, *An analytic construction of degenerating abelian varieties over complete rings*, Compositio Math. **24**, 239–272, 1972.

[OS91] K. Oguiso and T. Shioda, *The Mordell–Weil lattice of a rational elliptic surface*, Comment. Math. Univ. St. Pauli **40**, 83–99, 1991.

[Orl72] P. Orlik, *Seifert Manifolds*, Lecture Notes in Mathematics, vol. 291, Springer-Verlag, Berlin–New York, 1972.

[Ser90] F. Serrano, *Divisors of bielliptic surfaces and embeddings in $\mathbb{P}^4$*, Math. Z. **203**, 527–534, 1990.

[Shi90] T. Shioda, *On the Mordell–Weil lattices*, Comment. Math. Univ. St. Pauli **39**, 211–240, 1990.

[Sma61] S. Smale, *Generalized Poincaré's conjecture in dimensions greater than four*, Ann. of Math. (2) **74**, 391–406, 1961.

[Yau93] S.-T. Yau, *Open problems in geometry*, in *Differential Geometry: Partial Differential Equations on Manifolds* (Los Angeles, CA, 1990), Proc. Sympos. Pure Math., vol. 54, part 1, Amer. Math. Soc., Providence, RI, 1–28, 1993.